

Mild

1) List all the factors

- a) 24: 1, 2, 3, 4, 6, 8, 12, 24
 b) 36: 1, 2, 3, 4, 6, 9, 12, 18, 36
 c) 40: 1, 2, 4, 5, 8, 10, 20, 40
 d) 18: 1, 2, 3, 6, 9, 18

2) Circle the prime numbers

11, 17, 23, 29

3) Write the first five multiples

- a) of 6: 6, 12, 18, 24, 30
 b) of 9: 9, 18, 27, 36, 45
 c) of 4: 4, 8, 12, 16, 20

Spicy

1) Common factors and multiples

- a) 1, 2, 3, 4, 6, 12 (highest 12)
 b) LCM of 4 and 6: 12
 c) LCM of 6 and 8: 24

2) Sort into the diagram

Both (overlap): 4, 16, 36, 64

Square numbers only: 9, 25, 49

Multiples of 4 only: 20, 40

Outside both circles: none

3) True or false?

True. $6 = 2 \times 3$, so every multiple of 6 is also 3 times a whole number.

Hot

1) Prime factor trees

- a) $60 = 2 \times 2 \times 3 \times 5$
 b) $72 = 2 \times 2 \times 2 \times 3 \times 3$
 c) $84 = 2 \times 2 \times 3 \times 7$

2) Reasoning

- a) It is the square of a prime number, e.g. 4, 9, 25
 b) 1, 4, 9, 16, 25 (the square numbers)

Investigate

Layla is **right**. 16 has 5 factors (1, 2, 4, 8, 16), 25 has 3 (1, 5, 25) and 36 has 9. In a square number one factor pairs with itself ($4 \times 4 = 16$), so it is only counted once.